

Technology #15

MR Flow Quantification Sensor at Catheter
Scale

Engineering Technology-Transfer Dossier
CONFIDENTIAL

Technology Maturity: ENGINEERING_DEFINITION
Generated: 2026-08-27

Not physically validated. Not transfer-ready.

BUYER DECISION PAGE

WHAT IS IT?

MR Flow Quantification Sensor at Catheter Scale

WHY DOES IT MATTER?

CSF flow measurement requires clinical MRI (expensive, not continuous)

WHAT IS ACTUALLY ESTABLISHED?

Engineering definition with 5 governing equations, 12 design inputs, 9 failure modes. Computational model exists. No physical prototype.

WHAT IS NOT ESTABLISHED?

Physical validation, manufacturing process, regulatory pathway, IP ownership.

WHY MIGHT A BUYER CARE?

Opportunity in Magnetic Resonance (MRI/NMR) + Flow Quantification + Miniaturization. 8-step build plan with specific test articles and acceptance criteria. Target buyer: MRI/sensor company; neuroshunt manufacturer.

WHAT DOES THE BUYER GET?

MR flow quantification concept + mechanism description; Larmor + phase-contrast + SNR governing equations; Bench test protocols (SNR, flow accuracy, susceptibility, stability, EMC); External precedent catalog (miniaturized NMR, phase-contrast MRI); Critical UNKNOWN disclosure (miniaturization feasibility, SNR at catheter scale, susceptibility artifacts); This engineering dossier with failure analysis + build plan; Honest assessment: miniaturization to catheter scale may be physically blocked (high-risk concept)

WHAT DOES THE BUYER HAVE TO BUILD?

Production magnet design (B0, material, miniaturization); Production RF + gradient coil design; Production signal processing + flow calculation; Manufacturing process for catheter-scale MR sensor; Regulatory submission (likely Class III — novel MRI device); Clinical validation (flow accuracy vs standard); Miniaturization feasibility proof (may be blocked); MRI safety assessment + labeling

WHAT IS THE NEXT DECISIVE EXPERIMENT?

SNR feasibility test (magnet + coil + phantom)

WHAT WOULD MAKE US KILL IT?

Miniaturization to catheter scale cannot achieve $SNR > 10:1$ at any B0 field (physically blocked)

WHAT TRANSACTION COULD MAKE SENSE?

Sponsored Validation

What Exists Today

Existing:

- + Documented mechanism with governing equations
- + Computational model
- + 5 external evidence sources
- + Design inputs and outputs
- + Failure analysis
- + 8-step engineering build plan
- + Transfer boundary definition

Does Not Yet Exist:

- Physical prototype
- Qualified manufacturing process
- Clinical data
- Verified IP ownership
- Regulatory clearance
- Commercial product

Buyer Can Commission:

- > Specific bench experiment (see build plan)
- > Prototype fabrication
- > Materials study
- > Manufacturing feasibility study
- > Regulatory pre-submission

Development / De-risking Ladder

CURRENT STATE: ENGINEERING_DEFINITION

NEXT ENGINEERING STEP: SNR feasibility test (magnet + coil + phantom)

NEXT VALIDATION STEP: Bench prototype validation per build plan

NEXT REGULATORY STEP: Pre-submission to FDA for pathway classification

COMMERCIALIZATION GATE: Clinical validation evidence required

TRANSFER GATE: Transfer-ready when physical validation + regulatory pathway + IP verified

Engineering Investment Ladder

Next \$5K action: WP-01 - SNR feasibility test (magnet + coil + phantom) - 12 weeks

Basis: Build plan WP. Scope: SNR at target B0. Uncertainty: Effort estimate, not quoted.

Next \$25K action: WP-03 - Flow phantom with known flow rates - 10 weeks

Basis: Build plan WP. Scope: Flow measurement accuracy vs true flow. Uncertainty: Effort estimate, not quoted.

Next \$100K action: WP-05 - Magnet stability specimens - 12 weeks

Basis: Build plan WP. Scope: B0 drift vs temperature + time. Uncertainty: Effort estimate, not quoted.

Top 5 Buyer Diligence Questions

Q1: Is the mechanism physically credible for the intended application?

The governing model includes 5 equations. Key assumptions: CSF protons behave as in bulk water (valid approximation); B0 field stable over measurement period; Gradient coil produces linear gradient; RF coil sensitivity uniform over measurement volume; Phase wraparound managed (VENC chosen correctly); Motion artifacts (patient motion, physiologic motion) separable from flow. UNKNOWN: whether assumptions hold in vivo.

Q2: What evidence supports the core claim?

5 external evidence sources. All are external precedent, not physical validation of this specific invention.

Q3: What is the regulatory pathway?

UNKNOWN. Not yet determined. Requires pre-submission to FDA. No predicate identified for some packages.

Q4: Who owns the IP?

UNKNOWN. IP ownership not verified. Requires legal counsel engagement.

Q5: What would it cost to validate?

First experiment: SNR feasibility test (magnet + coil + phantom). Estimated effort from build plan. Full validation cost: COST TO BE QUOTED.

Commercial Interest

TARGET APPLICATION: CSF shunt systems for hydrocephalus management

CURRENT ALTERNATIVE: Standard fixed-pressure or programmable CSF shunt valves

MECHANISM DIFFERENCE: Miniaturized permanent magnet + RF coil + gradient coil enables phase-contrast MRI flow measurement at catheter scale

POTENTIAL VALUE DRIVER: Addresses: CSF flow measurement requires clinical MRI (expensive, not continuous)

ADOPTION BARRIER: Requires physical validation, regulatory clearance, and manufacturing qualification

DEVELOPMENT BARRIER: No physical prototype exists; manufacturing process not established

Licensee Capability Fit

IDEAL BUYER: MRI/sensor company; neuroshunt manufacturer

REQUIRED ENGINEERING CAPABILITY: Catheter design, fluid mechanics / domain-specific engineering

REQUIRED MANUFACTURING CAPABILITY: Medical tubing extrusion, injection molding, cleanroom assembly

REQUIRED REGULATORY CAPABILITY: FDA 510(k) or PMA submission experience

REQUIRED CAPITAL: Validation: \$50-200K. Full development: \$500K-2M (estimates, not quoted)

REQUIRED MARKET ACCESS: Neurosurgery market channels; hospital purchasing relationships

1. Technology Description

Domain: Magnetic Resonance (MRI/NMR) + Flow Quantification + Miniaturization

Disciplines: MRI/NMR physics, RF coil engineering (miniaturized), Signal processing (phase-contrast flow), Magnetic field engineering (portable magnets)

Architecture: Implantable MR flow sensor that quantifies CSF flow rate in shunt catheter using phase-contrast MRI principles. Miniaturized magnet + RF coil integrated into catheter or external patch.

Subsystems:

SS-01: Miniaturized permanent magnet (B0 source) - Provides static magnetic field B0 for nuclear spin polarization

SS-02: RF coil (transmit/receive) - Excites spins at Larmor frequency; receives echo signal

SS-03: Gradient coil (bipolar) - Applies bipolar gradient for phase-contrast velocity encoding

SS-04: Signal processing (phase extraction + flow calc) - Extracts phase shift; computes velocity from phase difference

2. Mechanism Architecture

Physical Changes: Protons in flowing CSF experience phase shift under bipolar gradient; static protons have zero net phase; flowing protons have phase proportional to velocity

Key Physics: Phase-contrast MRI: stationary tissue has zero phase accumulation under bipolar gradient; moving spins accumulate phase $d\phi = \gamma \cdot M1 \cdot v$. Two acquisitions (bipolar polarities) yield phase difference proportional to velocity.

3. Governing Engineering Model

Summary: Phase-contrast MRI flow measurement: bipolar gradient encodes velocity as phase shift; two acquisitions with opposite gradient polarities yield phase difference proportional to velocity.

Equations:

$\omega = \gamma \cdot B0$ [Larmor frequency for P-29 miniaturized magnet at catheter scale]

$d\phi = \gamma \cdot M1 \cdot v$ [phase encoding of CSF flow velocity in P-29 catheter-scale MR sensor]

$VENC = \pi / (\gamma \cdot M1)$ [velocity encoding]

$Q = v \cdot A$ [volumetric flow from velocity]

$SNR \sim B0 \cdot \sqrt{N \cdot voxel_volume} / \sqrt{BW}$ [SNR scaling]

Assumptions:

- CSF protons behave as in bulk water (valid approximation)
- B0 field stable over measurement period
- Gradient coil produces linear gradient
- RF coil sensitivity uniform over measurement volume
- Phase wraparound managed (VENC chosen correctly)
- Motion artifacts (patient motion, physiologic motion) separable from flow

Boundary Conditions:

- B0: 0.1-1 T typical for miniaturized permanent magnets (vs 1.5-3 T clinical MRI)
- Larmor frequency: 4-43 MHz at 0.1-1 T
- Flow velocity: 0.1-10 cm/s typical in shunt catheter
- Measurement volume: catheter lumen (~1-2 mm diameter, ~1-5 cm length)

Failure Regimes:

- B0 too low -> SNR insufficient for phase measurement

- Gradient coil too weak -> VENC too high for slow CSF flow
- RF coil sensitivity too low -> SNR insufficient
- Susceptibility artifacts (catheter material) -> phase errors
- Motion artifacts overwhelm flow signal
- Phase wraparound (velocity > VENC) -> ambiguous measurement

4. Design Inputs

ID	Input	Value	Origin
DI-001	Clinical need	Chronic CSF flow monitoring enables shunt function assessment; current standard requires imaging or invasive measurement	VERIFIED
DI-002	Functional requirement	MR flow quantification sensor using phase-contrast MRI principles at miniaturized scale	MODELLED
DI-003	Larmor frequency	EXTERNAL_PRECEDENT (42.58 MHz/T for proton)	EXTERNAL_PRECEDENT
DI-004	CSF flow range	EXTERNAL_PRECEDENT (~0.1-10 cm/s typical in shunt)	EXTERNAL_PRECEDENT
DI-005	B0 target	UNKNOWN — design choice (likely 0.1-1 T for miniaturization)	ENGINEERING_PROPOSED
DI-006	VENC target	MODELLED (matches CSF flow range ~1-10 cm/s)	MODELLED
DI-007	Form factor	UNKNOWN — implantable catheter-integrated or external patch	UNKNOWN
DI-008	Biocompatibility (ISO 10993)	UNKNOWN (for implantable components)	UNKNOWN
DI-009	Sterilization	UNKNOWN	UNKNOWN
DI-010	EMC	UNKNOWN — required IEC 60601-1-2	UNKNOWN
DI-011	MRI safety (if implantable magnet)	UNKNOWN — MRI-contraindicated labeling may be required	UNKNOWN
DI-012	Power budget	UNKNOWN	UNKNOWN

5. Design Outputs

ID	Description	Status	Missing Inputs
DO-001	Miniaturized magnet design (B0, geometry, material)	ABSENT	B0 target; Magnet material (NdFeB, SmCo); Geometry constraint
DO-002	RF coil design (frequency, geometry, sensitivity)	ABSENT	Larmor frequency (depends on B0); Coil geometry; Sensitivity target
DO-003	Gradient coil design (amplitude, linearity)	ABSENT	Gradient amplitude target; Coil geometry; Power budget
DO-004	Signal processing + flow calculation	MODELLED	

6. Critical Design Parameters

Name	Value	Unit	Basis	Verification
B0 field strength	UNKNOWN — design choice (likely 0.1-1 T for miniaturization)		Magnetization tradeoff	Field mapping
Gradient amplitude G	UNKNOWN	mT/m	Coil design + power budget	
Gradient duration (for M1)	UNKNOWN	ms	Sequence design	
VENC (velocity encoding)	UNKNOWN — target matches CSF/flow range (~1.0 m/s)		Flow range	
RF coil size	UNKNOWN — constrained by catheter ~2 mm or external patch		External patch	Geometric verification with catheter vendor
RF coil sensitivity	UNKNOWN	dimensionless	Coil design	
SNR	UNKNOWN — must be sufficient for phaseless demod		Derived from B0, coil, voxel, averaging	

7. Failure Modes

Mode	Mechanism	Mitigation	Verification	Residual Uncertainty
SNR insufficient at low B0	Unmagnetized magnet provides little offset (hard polarization at 0.1T); signal averaging (longer B0 acquisition) helps	High offset (hard polarization at 0.1T); signal averaging (longer B0 acquisition) helps	SNR, signal averaging (longer B0 acquisition) helps	UNKNOWN (likely) SNR at catheter scale
Gradient coil insufficient	Weak gradient can't encode strong flow	Stronger gradient + longer dwell	VENC measurement	UNKNOWN achievable gradient strength
Susceptibility artifacts	Catheter material distorts B0	MR-compatible catheter material	Phantom test with catheter	UNKNOWN artifact magnitude
Motion artifacts	Patient motion or physiologic motion corrupts phase	Gating + composite phase + averaging	Respiration/circulation test	UNKNOWN motion magnitude in vivo
Phase wraparound	Velocity > VENC causes ambiguity	Adaptive VENC; multi-VENC acquisition	Variable flow test	UNKNOWN flow range in vivo
Magnet instability (drift)	Permanent magnet field drifts	Temperature compensation; Temperature cycling	Temperature cycling	UNKNOWN drift rate
EMC interference	External EM affects RF measurement	Shielding + IEC 60601-1-2	EMC test	UNKNOWN interference environment
Power consumption (at standard RF pulse)	Standard RF pulses consume too much power	Ultra-low power design + duty cycling	Power budget test	UNKNOWN achievable battery life
Magnet safety (projection)	Permanent magnet attracts ferromagnetic objects	Shielding + MRI Safety guidelines	Safety assessment	UNKNOWN safety profile

8. Failure Analysis

Failure Mode	Mechanism	Mitigation	Residual Uncertainty
SNR insufficient at low B0	Miniaturized magnet provides insufficient B0	Higher B0 (harder to miniaturize); averaging	UNKNOWN achievable SNR
Gradient coil insufficient	Weak gradient can't encode slow flow	Stronger gradient + longer duration	UNKNOWN achievable gradient
Susceptibility artifacts	Catheter material distorts B0	MR-compatible materials + correction	UNKNOWN artifact magnitude
Motion artifacts	Patient/physiologic motion corrupts phase	gating + motion correction + averaging	UNKNOWN motion magnitude
Phase wraparound	Velocity > VENC	Adaptive VENC; multi-VENC	UNKNOWN flow range
Magnet drift	Permanent magnet field drifts with temperature	Temperature compensation + monitoring	UNKNOWN drift rate
EMC interference	External EM affects RF	Shielding + IEC 60601-1-2	UNKNOWN interference environment
Power consumption	Gradient + RF consume power	Low-power design + duty cycling + energy harvesting	UNKNOWN achievable battery life
Magnet safety (projectiles)	Permanent magnet attracts ferromagnetic objects	Shielding + warnings + MRI-contraindications	UNKNOWN safety profile

9. Verification Strategy

ID	Requirement	Method	Acceptance	Result
V-001	Flow measurement accuracy in phantom	Phantom with known flow	~10-20% accuracy	NOT_TESTED
V-002	SNR > threshold	Bench SNR measurement	SNR > 10:1	NOT_TESTED
V-003	VENC appropriate	Variable flow test	VENC matches flow range	NOT_TESTED
V-004	Susceptibility artifact management	Phantom with catheter	Artifacts correctable	NOT_TESTED
V-005	Biocompatibility ISO 10993	ISO 10993	Pass	NOT_TESTED
V-006	EMC IEC 60601-1-2	EMC test	Pass	NOT_TESTED

10. Validation Strategy

ID	Requirement	Method	Acceptance	Result
VAL-001	Clinical flow measurement accuracy	Clinical study vs standard flow measurement	Clinically useful accuracy	NOT_PERFORMED

11. Materials

Component	Candidate Material	Source	Verification Required	Status
Permanent magnet	NdFeB (neodymium iron boron)	Standard high-field permanent magnet supplier	Field stability + temperature coefficient	CANDIDATE — note: iron plating
Permanent magnet	SmCo (samarium cobalt)	High-temperature-stable permanent magnet supplier	Field stability + biocompatibility	CANDIDATE — preferred for temperature stability
RF coil	Copper wire / trace	Standard RF coil material	Biocompatibility (if implantable) + impedance	CANDIDATE
Housing (if implantable)	Titanium or polymer	Standard implantable housing	ISO 10993 + hermeticity + MR compatibility	CANDIDATE — note: metal housing affects field

12. Bill of Materials

Item	Description	Qty	Type	Material	Supplier	Criticality	Verification
01	Miniaturized permanent magnet (B0.5T)	1	CUSTOM_COMPONENT	NdFeB or SmCo (rare earth magnets)	Multiple magnet manufacturers	CRITICAL	Field mapping + stability + safety
02	RF coil (transmit/receive)	1	CUSTOM_COMPONENT	Copper or silver wire / PCB trace	PCB manufacturer — custom	CRITICAL	Calibration + tuning + matching + sensitivity
03	Gradient coil	1	CUSTOM_COMPONENT	Copper wire / PCB trace	UNKNOWN — custom	CRITICAL	Linearity + amplitude
04	RF transceiver + signal processor	1	COTS_CANDIDATE	IC (COTS or custom)	Multiple	HIGH	Calibration + EMC
05	Power supply (battery or energy harvester)	1	CUSTOM_COMPONENT	Rechargeable battery or piezoelectric harvester	Manufacturer (P-15)	HIGH	Lifetime + safety

13. Manufacturing

Status: ENGINEERING_CANDIDATE — no catheter-scale MR flow sensor process validated. Miniaturization feasibility is a critical UNKNOWN.

Candidate Processes:

- Magnet machining + magnetization: Standard magnet manufacturing - Custom for catheter-scale size
- RF coil winding or PCB fabrication: Standard coil manufacturing - Custom for catheter-scale size
- Hybrid assembly (magnet + coils + electronics): Standard hybrid assembly - Miniaturization required
- Hermetic encapsulation (if implantable): Standard implantable packaging - Tradeoff between encapsulation and MR performance

14. External Evidence

Source 1: A miniaturized spectrometer for NMR relaxometry under

URL: <https://www.nature.com/articles/s41598-019-47634-2>

Excerpt: by Y Tang · 2019 · Cited by 24 — In this work, we extend NMR relaxometry on a minute sample below 20 μ L to challenging environment of 150 °C in temperature and 900 bar in ...

Source 2: μ -NMR Technology for Biomedical Applications: A Review

URL: <https://www.mdpi.com/2227-9040/12/12/248>

Excerpt: by B Sequeira-Antunes · 2024 · Cited by 5 — By miniaturizing NMR systems, micro- NMR (μ - NMR) devices overcome the limitations of traditional bulky NMR instruments , making them more ...

Source 3: (PDF) Miniaturised NMR sensor

URL: https://www.researchgate.net/publication/262183456_Miniaturised_NMR_sensor

Excerpt: Realisation of a miniaturised NMR sensor | ... picoSpin-45 proton NMR spectrometer that weight less than 5 kg costing around \$20,000. Such a device is able to ...

Source 4: Miniaturized, biopsy-implantable chemical sensor with

URL: <https://pubmed.ncbi.nlm.nih.gov/26177607>

Excerpt: by CC Vassiliou · 2015 · Cited by 21 — Here we demonstrate a miniaturized wireless sensor that can be implanted during a biopsy procedure and return chemical information from within ... Read more

Source 5: Sensors for medical devices | IST AG

URL: <https://innuscreen.ist-ag.com/en/medical-devices>

Excerpt: Biosensors, flow , temperature or conductivity sensors combined with viral RNA extraction kits complete IST AG's medical product portfolio.

15. Transfer Boundary

YOU RECEIVE:

- + MR flow quantification concept + mechanism description
- + Larmor + phase-contrast + SNR governing equations
- + Bench test protocols (SNR, flow accuracy, susceptibility, stability, EMC)
- + External precedent catalog (miniaturized NMR, phase-contrast MRI)
- + Critical UNKNOWN disclosure (miniaturization feasibility, SNR at catheter scale, susceptibility artifacts)
- + This engineering dossier with failure analysis + build plan
- + Honest assessment: miniaturization to catheter scale may be physically blocked (high-risk concept)

YOU MUST DEVELOP:

- > Production magnet design (B0, material, miniaturization)
- > Production RF + gradient coil design
- > Production signal processing + flow calculation
- > Manufacturing process for catheter-scale MR sensor
- > Regulatory submission (likely Class III — novel MRI device)
- > Clinical validation (flow accuracy vs standard)
- > Miniaturization feasibility proof (may be blocked)
- > MRI safety assessment + labeling

YOU MUST VERIFY:

- ? Biocompatibility (ISO 10993)
- ? Sterilization compatibility
- ? Mechanical integrity
- ? Regulatory pathway
- ? IP ownership and freedom to operate
- ? Manufacturing process capability

16. Open Questions

- Achievable SNR at catheter-scale B0 — UNKNOWN (critical for feasibility)
- Miniaturization limit for magnet + coils — UNKNOWN
- Susceptibility artifact magnitude — UNKNOWN
- Motion artifact magnitude in vivo — UNKNOWN
- Magnet stability over implantation lifetime — UNKNOWN
- Power budget for active operation — UNKNOWN
- Regulatory pathway (likely Class III — novel MRI device) — UNKNOWN
- Clinical accuracy requirement for shunt flow monitoring — UNKNOWN
- Whether concept is physically plausible at device scale — UNKNOWN (may be blocked by miniaturization)

17. Decisive Next Experiment

WP	Test Article	Equipment	Measurement	Acceptance	Deliverable	Effort
WP-01	SNR feasibility test (miniaturized NMR setup)	Bench NMR setup	SNR at target B0	SNR > 10:1 sufficient	SNR feasibility report	10 weeks

WP	Test Article	Equipment	Measurement	Acceptance	Deliverable	Effort
WP-02	Gradient coil prototype	Bench gradient characterization	Gradient amplitude + linearity	Gradient strength sufficient for 1.5T	Gradient map	10 weeks
WP-03	Flow phantom with known flow rates	MR flow measurement system	Flow measurement accuracy	Flow accuracy within 10-20%	Flow accuracy report	10 weeks
WP-04	Susceptibility artifact phantom with catheters	Phantom with catheters	Phantom vs material	Artifacts within correction	Artifacts report	8 weeks
WP-05	Magnet stability specification	Temperature chamber	Field vs temperature	Drift $\pm$ 10 ppm over operation	Stability report	12 weeks
WP-06	Biocompatibility specification	ISO 10993 (table)	ISO 10993 series	Pass	ISO 10993 report	12 weeks (external)
WP-07	EMC test articles (miniaturized magnet + receiver)	EMC test magnet + receiver	EMC emissions + immunity	Pass (receiver)	EMC report	8 weeks (external)
WP-08	Miniaturization feasibility study	Prototype, characterization	Characterization	Characterization	Miniaturization report	15 weeks (high risk)

18. Buyer Decision Framework

WHY INVEST: Engineering definition exists with 5 governing equations and a 8-step build plan. The next experiment is clearly specified. Target buyer: MRI/sensor company; neuroshunt manufacturer.

WHY WAIT: No physical prototype exists. Regulatory pathway is unresolved. IP ownership is unverified. Manufacturing process is not qualified.

WHY REJECT: Miniaturization to catheter scale cannot achieve SNR > 10:1 at any B0 field (physically blocked)

WHAT WOULD CHANGE THE DECISION: A successful bench experiment demonstrating the core mechanism. A qualified regulatory pathway. A confirmed IP position.

19. Kill Condition

KILL_IF: Miniaturization to catheter scale cannot achieve SNR > 10:1 at any B0 field (physically blocked)

DISCLOSURE

This dossier distinguishes established evidence, external precedent, computational/modelled results, engineering proposals, and unresolved questions. The absence of physical validation should not be interpreted as evidence that the technology will fail; it means the relevant proposition has not yet been established experimentally.

This dossier does not constitute legal, patentability, FTO, regulatory, or investment advice.